

CUXHAVEN, Germany

WMO: 101310

Lat: 53.8714N

Lon: 8.7058E

Elev: 39

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.0	20.7	11.8	10.0	18.8	15.1	11.9	21.6	35.2	40.0	32.4	35.8	13.2	90	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	77.4	65.9	74.0	64.7	70.9	63.2	67.4	74.5	65.8	72.0	64.3	69.6	12.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	91.8	70.9	63.3	87.1	69.3	61.7	82.4	67.6	31.8	74.8	30.6	72.0	29.4	69.6	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.8	26.4	23.6	DB	17.3	85.7	7.9	5.1	11.6	89.4	7.0	92.3	2.5	95.2	-3.2	98.8	(4)
(5)			WB	16.2	72.1	7.4	1.7	10.9	73.4	6.6	74.4	2.5	75.3	-2.9	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	35.4	35.0	40.1	45.7	53.8	59.0	62.8	62.6	57.7	51.1	42.7	38.5		
	DBStd	11.49	8.14	7.26	5.79	5.15	5.89	5.50	4.56	4.40	3.66	5.19	5.40	6.11		
	HDD50	1987	451	422	309	149	25	1	0	0	0	49	223	358		
	HDD65	6005	916	841	772	580	352	197	100	102	222	431	670	822		
	CDD50	1541	0	1	3	19	141	270	397	392	230	83	4	1		
	CDD65	83	0	0	0	0	4	16	31	29	3	0	0	0		
	CDH74	302	0	0	0	1	20	49	105	119	8	0	0	0		
	CDH80	43	0	0	0	0	1	4	14	24	0	0	0	0		
	WSAvg	12.4	14.3	13.8	13.0	11.9	12.0	11.3	11.4	10.9	11.7	12.8	12.9	13.5		
	PrecAvg	33.6	2.6	2.0	2.0	1.6	2.2	3.3	3.6	3.6	3.3	3.2	3.1	3.0		
(15) Precipitation	PrecMax	44.9	5.5	4.6	3.7	3.8	4.6	7.2	7.2	8.5	7.9	7.9	6.5	7.4		
	PrecMin	19.6	0.0	0.7	0.4	0.2	0.5	0.3	1.5	1.1	1.1	1.1	0.4	0.7		
	PrecStd	5.6	1.3	1.0	1.0	0.8	1.0	1.7	1.7	1.8	1.8	1.7	1.5	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.5	55.9	59.6	65.3	76.4	79.3	81.5	82.7	74.3	67.5	56.4	53.5		
		MCWB	48.8	51.4	51.7	54.5	62.5	66.8	68.4	68.8	64.8	60.8	52.5	49.9		
	2%	DB	48.5	50.7	52.7	60.4	71.4	74.6	76.8	77.0	68.6	63.5	53.6	50.8		
		MCWB	46.5	47.0	48.4	51.8	60.9	64.3	66.5	66.2	61.3	58.1	51.2	48.6		
	5%	DB	46.6	46.8	49.8	56.6	66.7	70.7	73.2	72.8	65.5	60.1	51.7	48.5		
		MCWB	44.9	44.0	46.2	50.6	58.4	62.6	65.0	64.6	60.3	56.2	49.8	46.6		
	10%	DB	44.9	44.0	47.5	53.6	62.7	67.4	69.9	69.3	63.3	57.6	50.0	46.5		
		MCWB	43.1	41.6	44.8	48.8	56.4	61.1	63.4	63.1	58.6	54.5	48.1	45.0		
	0.4%	WB	49.3	52.1	52.6	56.6	63.7	68.8	69.9	69.8	65.8	62.3	53.6	50.9		
		MCDB	50.7	55.7	58.8	62.8	73.3	76.7	77.3	78.6	71.0	66.6	55.3	52.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.7	47.4	48.8	53.1	61.4	65.5	67.5	67.2	63.2	59.0	51.5	48.9		
		MCDB	47.8	50.1	51.6	58.1	70.2	71.2	73.8	75.0	67.4	62.2	53.2	50.8		
	5%	WB	45.2	44.3	46.8	51.2	59.1	63.6	65.9	65.6	61.1	56.8	50.0	46.9		
		MCDB	46.5	46.8	49.2	55.9	65.7	69.1	71.9	71.8	64.5	59.4	51.5	48.2		
	10%	WB	43.4	41.9	45.1	49.2	56.8	61.8	64.2	63.9	59.5	55.2	48.4	45.1		
		MCDB	44.8	43.7	47.2	53.0	62.3	67.3	69.5	68.8	62.3	57.1	49.9	46.3		
	5% DB	MDBR	6.5	7.0	8.4	10.2	10.6	9.4	9.7	10.1	8.9	8.3	7.0	6.2		
		MCDBR	7.1	10.3	12.2	16.2	17.9	15.8	16.0	17.5	13.6	12.6	9.1	8.2		
		MCWBR	6.7	8.2	8.9	10.2	10.2	9.2	8.8	9.4	8.7	8.5	8.0	7.5		
		MCWBR	7.1	10.0	11.4	14.7	16.9	13.8	14.3	14.6	11.8	10.4	8.8	7.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.8	8.5	9.1	10.0	10.0	9.0	9.0	8.7	8.9	8.0	8.0	7.4		
	taub		0.328	0.346	0.375	0.408	0.401	0.400	0.416	0.411	0.387	0.363	0.341	0.319		
		taud	2.309	2.326	2.270	2.236	2.306	2.334	2.302	2.326	2.378	2.398	2.348	2.304		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		197	235	252	259	268	270	262	256	247	226	190	174		
		Edh at Noon	19	25	33	39	38	37	38	35	30	23	18	16		
	RadAvg		174	384	772	1282	1610	1677	1593	1311	944	519	218	130		
(45) All-Sky Solar Radiation	RadStd		20	53	75	150	170	136	203	98	109	57	23	10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-238	N/A	N/A

Nomenclature: See separate page